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SCALE FOR PROJECT READY SET BOOLE (/PROJECTS/READY-SET-BOOLE)

You should evaluate 1 student in this team



Git repository

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Introduction

- Remain polite, courteous, respectful and constructive throughout the evaluation process. The well-being of the community depends on it.
- Identify with the person (or the group) evaluated the eventual dysfunctions of the work. Take the time to discuss and debate the problems you have identified.
- You must consider that there might be some difference in how your peers might have understood the project's instructions and the scope of its functionalities. Always keep an open mind and grade him/her as honestly as possible. The pedagogy is valid only and only if peer-evaluation is conducted seriously.

Guidelines

- Only grade the work that is in the student or group's GiT repository.
- Double-check that the GiT repository belongs to the student or the group. Ensure that the work is for the relevant project and also check that "git clone" is used in an empty folder.
- Check carefully that no malicious aliases was used to fool you and make you evaluate something other than the content of the official repository.
- To avoid any surprises, carefully check that both the evaluating and the evaluated students have reviewed the possible scripts used to facilitate the grading.
- If the evaluating student has not completed that particular project yet, it is mandatory for this student to read the entire subject prior to starting the defence.
- Use the flags available on this scale to signal an empty repository, non-functioning program, a norm error, cheating etc. In these cases, the grading is over and the final grade is 0 (or -42 in case of cheating). However, with the exception of cheating, you are

encouraged to continue to discuss your work (even if you have not finished it) in order to identify any issues that may have caused this failure and avoid repeating the same mistake in the future.

- Remember that for the duration of the defence, no segfault, no other unexpected, premature, uncontrolled or unexpected termination of the program, else the final grade is 0. Use the appropriate flag.

You should never have to edit any file except the configuration file if it exists.

If you want to edit a file, take the time to explicit the reasons with the evaluated student and make sure both of you are okay with this.

- You must also verify the absence of memory leaks. Any memory allocated on the heap must be properly freed before the end of execution.

You are allowed to use any of the different tools available on the computer, such as leaks, valgrind, or e_fence. In case of memory leaks, tick the appropriate flag.

Attachments

 subject.pdf (<https://cdn.intra.42.fr/pdf/pdf/185035/en.subject.pdf>)

Exercise 00 - Adder

Complexity

Ask the student to justify the complexity of the function. It must be at most $O(\log n)$ in time and $O(\log n)$ in space.

☒ Yes

☐ No

Used operators

Check that the only operators that were used in the function are:

- & (bitwise AND)
- | (bitwise OR)
- ^ (bitwise XOR)
- << (left shift)
- >> (right shift)
- = (assignment)
- ==, !=, <, >, <=, >= (comparison operators)
- The increment operator (only to increment the index of a loop)

Check for the use of any forbidden mathematical functions (see the subject).

☒ Yes

☐ No

Basic tests

Check the behaviour of the function with the following parameters:

- 'adder(0, 0)' gives '0'
- 'adder(1, 0)' gives '1'
- 'adder(0, 1)' gives '1'
- 'adder(1, 1)' gives '2'
- 'adder(1, 2)' gives '3'

- 'adder(2, 2)' gives '4'

Feel free to perform more tests on your own.

✓ Yes

✗ No

Exercise 01 - Multiplier

Complexity

Ask the student to justify the complexity of the function. It must be at most $O(1)$ in time and $O(1)$ in space.

✓ Yes

✗ No

Used operators

Check that the only operators that were used in the function are:

- & (bitwise AND)
- | (bitwise OR)
- ^ (bitwise XOR)
- << (left shift)
- >> (right shift)
- = (assignment)
- ==, !=, <, >, <=, >= (comparison operators)
- The increment operator (only to increment the index of a loop)

Check for the use of any forbidden mathematical functions (see the subject).

✓ Yes

✗ No

Basic tests

Check the behaviour of the function with the following parameters:

- 'multiplier(0, 0)' gives '0'
- 'multiplier(1, 0)' gives '0'
- 'multiplier(0, 1)' gives '0'
- 'multiplier(1, 1)' gives '1'
- 'multiplier(1, 2)' gives '2'
- 'multiplier(2, 2)' gives '4'

Feel free to perform more tests on your own.

✓ Yes

✗ No

Exercise 02 - Gray code

Basic tests

Check the behaviour of the function. The binary representation of the returned number must correspond to the encoding of the given parameter in Gray code.

You can use an online Binary -> Gray code converter to make evaluation

easier.

Check for the use of any forbidden mathematical functions (see the subject).

☒ Yes

☐ No

Exercise 03 - Boolean evaluation

Complexity

Ask the student to justify the complexity of the function. It must be at most $O(n)$ in time.

☒ Yes

☐ No

Basic tests

Check the behaviour of the function with the following formulas:

- '0!' gives 'true'
- '1!' gives 'false'
- '00!' gives 'false'
- '10!' gives 'true'
- '01!' gives 'true'
- '11!' gives 'true'
- '10&' gives 'false'
- '11&' gives 'true'
- '11^' gives 'false'
- '10^' gives 'true'
- '00>' gives 'true'
- '01>' gives 'true'
- '10>' gives 'false'
- '11>' gives 'true'
- '00=' gives 'true'
- '11=' gives 'true'
- '10=' gives 'false'
- '01=' gives 'false'

Feel free to perform more tests on your own

Check for the use of any forbidden mathematical functions (see the subject).

☒ Yes

☐ No

Composition

Check the behaviour of the function with the following formulas:

- '11&0!' gives 'true'
- '10&1!' gives 'true'
- '11&1!' gives 'true'
- '11&1|1^' gives 'false'
- '01&1|1=' gives 'true'
- '01&1&1&' gives 'false'
- '0111&&&' gives 'false'

Feel free to perform more tests on your own.

✓ Yes

✗ No

Exercise 04 - Truth table

Complexity

Ask the student to justify the complexity of the logic part of the assignment (not the display of the result). Most $O(2^n)$ in time.

✓ Yes

✗ No

Basic tests

Check the behaviour of the function with the following formulas:

- 'A' must print:

```
,  
| A | |  
|---|---|  
| 0 | 0 |  
| 1 | 1 |  
,
```

- 'A!' must print:

```
,  
| A | |  
|---|---|  
| 0 | 1 |  
| 1 | 0 |  
,
```

- 'AB!' must print:

```
,  
| A | B | |  
|---|---|---|  
| 0 | 0 | 0 |  
| 0 | 1 | 1 |  
| 1 | 0 | 1 |  
| 1 | 1 | 1 |  
,
```

- 'AB&' must print:

```
,  
| A | B | |  
|---|---|---|  
| 0 | 0 | 0 |  
| 0 | 1 | 0 |  
| 1 | 0 | 0 |  
| 1 | 1 | 1 |  
,
```

- 'AB^' must print:

```
,  
| A | B | |  
|---|---|---|  
| 0 | 0 | 0 |  
| 0 | 1 | 1 |
```

```

| 1 | 0 | 1 |
| 1 | 1 | 0 |
,

```

- 'AB>' must print:

```

,
| A | B | |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |
,

```

- 'AB=' must print:

```

,
| A | B | |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |
,

```

- 'AA=' must print:

```

,
| A | |
|---|---|
| 0 | 1 |
| 1 | 1 |
,

```

Feel free to perform more tests on your own.

Check for the use of any forbidden mathematical functions (see the subject).

☒ Yes

☐ No

Composition

Check the behaviour of the function with the following formulas:

- 'ABC==' must print:

```

,
| A | B | C | |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |
,

```

- 'AB>C>' must print:

```

,
| A | B | C | |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |

```

```

| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

```

- 'AB>A>A>' must print:

```

| A | B | |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

```

By the way, the last formula is called Pierce's Law. You may want to check out what this is if you want to go deeper in mathematical logic.

Feel free to perform more tests on your own.

☒ Yes

☐ No

Exercise 05 - Negation Normal Form

Basic tests

Check the behaviour of the function with the following formulas:

- 'A'
- 'A!'
- 'AB&!'
- 'AB|!'
- 'AB>!'
- 'AB=!'

For each case, every occurrence of '!' must be placed after a variable, and the truth table must be the same.

Feel free to perform more tests on your own.

Check for the use of any forbidden mathematical functions (see the subject).

☒ Yes

☐ No

Composition

Check the behaviour of the function with the following formulas:

- 'ABC||'
- 'ABC||!'
- 'ABC|&'
- 'ABC&|'
- 'ABC&|!'
- 'ABC^^'
- 'ABC>>'

For each case, every occurrence of '!' must be placed after a variable, and the truth table must be the same.

Feel free to perform more tests on your own.

☒ Yes

☐ No

Exercise 06 - Conjunctive Normal Form

Basic tests

Check the behaviour of the function with the following formulas:

- 'A'
- 'A!'
- 'AB&!'
- 'AB|!'
- 'AB>!'
- 'AB=!'

For each case, every occurrence of '!' must be placed after a variable, every conjunction must be located at the end of the formula, and the truth table must be the same.

Feel free to perform more tests on your own

Check for the use of any forbidden mathematical functions (see the subject).

☒ Yes

☐ No

Composition

Check the behaviour of the function with the following formulas:

- 'ABC||'
- 'ABC|||!'
- 'ABC|&'
- 'ABC&|'
- 'ABC&|!'
- 'ABC^^'
- 'ABC>>'

For each case, every occurrence of '!' must be placed after a variable, every conjunction must be located at the end of the formula, and the truth table must be the same.

Feel free to perform more tests on your own.

☒ Yes

☐ No

Exercise 07 - SAT

Complexity

Ask the student to justify the complexity of the function. It must be at most $O(2^n)$ in time.

☒ Yes☐ No**Basic tests**

Check the behaviour of the function with the following formulas:

- 'A' gives 'true'
- 'A!' gives 'true'
- 'AA|' gives 'true'
- 'AA&' gives 'true'
- 'AA!&' gives 'false'
- 'AA^' gives 'false'
- 'AB^' gives 'true'
- 'AB=' gives 'true'
- 'AA>' gives 'true'
- 'AA!>' gives 'true'

Feel free to perform more tests on your own. For each case, every occurrence of '!' must be placed after a variable, every conjunction must be located at the end of the formula, and the truth table must be the same.

Check for the use of any forbidden mathematical functions (see the subject).

☒ Yes☐ No**Composition**

Check the behaviour of the function with the following formulas:

- 'ABC||' gives 'true'
- 'AB&A!B!&&' gives 'false'
- 'ABCDE&&&&' gives 'true'
- 'AAA^^' gives 'true'
- 'ABCDE^^^^' gives 'true'

Feel free to perform more tests on your own. For each case, every occurrence of '!' must be placed after a variable, every conjunction must be located at the end of the formula, and the truth table must be the same.

☒ Yes☐ No

Exercise 08 - Powerset

Complexity

Ask the student to justify the complexity of the function. It must be at most $O(2^n)$ in time and space.

☒ Yes☐ No**Basic tests**

Check the function's behaviour with different sets. Each times, the number of subsets in the resulting powerset must be equal 2^n where 'n' is the length of the set.

The order of the elements in the returned array doesn't matter.

Try the following:

- '[]' gives ' [[]]' (1 subset)
- '[0]' gives ' [[], [0]]' (2 subset)
- '[0, 1]' gives ' [[], [0], [1], [0, 1]]' (4 subset)
- '[0, 1, 2]' gives ' [[], [0], [1], [2], [0, 1], [1, 2], [0, 2], [0, 1, 2]]' (8 subset)

Feel free to perform more tests on your own.

Check for the use of any forbidden mathematical functions (see the subject).

 Yes

 No

Exercise 09 - Set evaluation

Basic tests

Try the following:

- 'A' with ' [[]]', the function must return '[]'
- 'A!' with ' [[]]', the function must return '[]'
- 'A' with ' [[]42]]', the function must return '[42]'
- 'A!' with ' [[]42]]', the function must return '[]'
- 'A!B&' with ' [[]1, 2, 3], [2, 3, 4]]' the function must return '[4]'
- 'AB!' with ' [[]0, 1, 2], [[]]', the function must return '[0, 1, 2]'
- 'AB&' with ' [[]0, 1, 2], [[]]', the function must return '[]'
- 'AB&' with ' [[]0, 1, 2], [0]]', the function must return '[0]'
- 'AB&' with ' [[]0, 1, 2], [42]]', the function must return '[]'
- 'AB^' with ' [[]0, 1, 2], [0]]', the function must return '[1, 2]'
- 'AB>' with ' [[]0], [1, 2]]', the function must return '[1, 2]'
- 'AB>' with ' [[]0], [0, 1, 2]]', the function must return '[0, 1, 2]'

Feel free to perform more tests on your own.

Check for the use of any forbidden mathematical functions (see the subject).

 Yes

 No

Composition

Try the following:

- 'ABC|' with ' [[], [], [[]]', the function must return '[]'
- 'ABC|' with ' [[]0], [1], [2]]', the function must return '[0, 1, 2]'
- 'ABC|' with ' [[]0], [0], [0]]', the function must return '[0]'
- 'ABC&&' with ' [[]0], [0], [[]]', the function must return '[]'
- 'ABC&&' with ' [[]0], [0], [0]]', the function must return '[0]'
- 'ABC^^' with ' [[]0], [0], [0]]', the function must return '[0]'
- 'ABC>>' with ' [[]0], [0], [0]]', the function must return '[0]'

Feel free to perform more tests on your own.

 Yes

 No

Exercise 10 - Curve

Basic tests

Ask the student to explain why is a space filling curve continuous and why is his implementation continuous.

Try passing pairs of values to the function. For each unique pair of values, the function must return a unique value between 0 and 1 (included).

Check for the use of any forbidden mathematical functions (see the subject).

✓ Yes

✗ No

Exercise 11 - Inverse function

Basic tests

Use the previous function to test this one. Try executing the function 'inverse_map(map(x, y))'. For every pair of values 'x' and 'y', this function must return the exact same value.

Check for the use of any forbidden mathematical functions (see the subject).

✓ Yes

✗ No

Ratings

Don't forget to check the flag corresponding to the defense

✓ Ok

★ Outstanding project

Empty work

📄 Incomplete work

⚙️ Invalid compilation

📖 Cheat

💣 Crash

💧 Leaks

🚫 Forbidden function

💬 Can't support / explain code

Conclusion

Leave a comment on this evaluation (2048 chars max)

Finish evaluation

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